Tutorial No : 2

Tutorial Title: Perform recursive binary search using C.
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Program:

```
#include <stdio.h>
int binarySearch(int arr[], int low, int high, int key)
{
        if (low > high)
        return -1;

        int mid = (low + high) / 2;

        if(arr[mid]==key)
        return mid;

        if(key<arr[mid])
        return binarySearch(arr, low, mid - 1, key);

        else
        return binarySearch(arr, mid + 1, high, key);
}

int main()
{
        int n, key, i;

        printf("Enter the number of elements: ");
        scanf("%d", &n);

        int arr[n];

        printf("Enter the elements in sorted order:\n");
        for (i = 0; i < n; i++)
        scanf("%d", &arr[i]);
```

```
    printf("Enter the element to search: ");
    scanf("%d", &key);

    int result = binarySearch(arr, 0, n - 1, key);

    if (result != -1)
    printf("Element found at index %d", result);
    else
    printf("Element not found");

    return 0;
}
```

Output:

